

Network Scanning with Nmap - Report

 **Task1:** Network Scanning with Nmap

 **Deadline:** 12th Feb, 11:59 PM

 **Objective:** Learn how to scan a network and identify open ports.

What is Network Scanning?

Network scanning is a process used to identify active devices, open ports, and services running on a network. It is commonly used for security assessments, vulnerability detection, and network management. By scanning a network, administrators and security professionals can gather information about potential weaknesses and take steps to enhance security.

What is Nmap?

Nmap (Network Mapper) is a free and open-source tool used for network discovery and security auditing. It is widely used to scan networks, detect open ports, identify running services, and find vulnerabilities. Nmap supports different types of scans, including basic network scans, service version detection, OS fingerprinting, and aggressive scans for in-depth analysis.

Key Features:

- Host discovery
 - Port scanning
 - Service and version detection
 - OS detection
 - Vulnerability detection
-

Network Scanning Process

Basic Scan:

 **Command used:** `nmap <IP>`

- Identified active hosts.
- Detected open ports on the target system.

```

Nmap done: 1 IP address (1 host up) scanned in 32.87 seconds

C:\Windows\System32>nmap -O [redacted]
Starting Nmap 7.95 ( https://nmap.org ) at 2025-02-13 23:58 India Standard Time
Nmap scan report for 192.168.1.100
Host is up (0.00064s latency).
Not shown: 996 closed tcp ports (reset)
PORT      STATE SERVICE
135/tcp    open  msrpc
139/tcp    open  netbios-ssn
445/tcp    open  microsoft-ds
3306/tcp   open  mysql
No exact OS matches for host (If you know what OS is running on it, see https://nmap.org/submit/ ).
TCP/IP fingerprint:
OS:SCAN(V=7.95%E=4%D=2/13%OT=135%CT=1%CU=30350%PV=Y%DS=0%DC=L%G=Y%TM=67AE39
OS:E1P=1686-pc-windows-windows)SEQ(SP=102%GCD=1%ISR=109%TI=I%CI=I%II=I%SS=
OS:S%TS=A)SEQ(SP=103%GCD=1%ISR=107%TI=I%CI=I%II=I%SS=5%TS=A)SEQ(SP=105%GCD=
OS:1%ISR=108%TI=I%CI=I%II=I%SS=5%TS=A)SEQ(SP=105%GCD=3%ISR=100%TI=I%CI=I%II
OS:=I%SS=5%TS=A)SEQ(SP=107%GCD=1%ISR=108%TI=I%CI=I%II=I%SS=5%TS=A)OPS(O1=MF
OS:FD7NW8ST11%O2=MF7D7NW8ST11%O3=MF7D7NW8NNT11%O4=MF7D7NW8ST11%O5=MF7D7NW8S
OS:T11%O6=MF7D7ST11)WIN(W1=FFFF%W2=FFFF%W3=FFFF%W4=FFFF%W5=FFFF%W6=FFFF)ECN
OS:(R=Y%DF=Y%T=80%W=FFFF%O=MF7D7NW8NNS%CC=N%Q=)T1(R=Y%DF=Y%T=80%S=0%A=S+%F=
OS:AS%RD=0%Q=)T2(R=Y%DF=Y%T=80%W=0%S=Z%A=S+F=AR%O=%RD=0%Q=)T3(R=Y%DF=Y%T=80
OS:%W=0%S=Z%A=O%F=AR%O=%RD=0%Q=)T4(R=Y%DF=Y%T=80%W=0%S=A%A=O%F=R%O=%RD=0%Q=
OS:)T5(R=Y%DF=Y%T=80%W=0%S=Z%A=S+F=AR%O=%RD=0%Q=)T6(R=Y%DF=Y%T=80%W=0%S=A%
OS:A=O%F=R%O=%RD=0%Q=)T7(R=Y%DF=Y%T=80%W=0%S=Z%A=S+F=AR%O=%RD=0%Q=)U1(R=Y%
OS:DF=N%T=80%IPL=164%UN=0%RIPL=G%RID=G%RIPCK=Z%RUCK=G%RUD=G)IE(R=Y%DFI=N%T=
OS:80%CD=Z)

```

Service Detection Scan:

 **Command used:** `nmap -sV <IP>`

Findings:

- Detected services running on open ports.
- Identified service versions.

```

C:\Windows\System32>nmap -sV 192.168.1.100
Starting Nmap 7.95 ( https://nmap.org ) at 2025-02-13 23:51 India Standard Time
Nmap scan report for 192.168.1.100
Host is up (0.00010s latency).
Not shown: 996 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
135/tcp    open  msrpc        Microsoft Windows RPC
139/tcp    open  netbios-ssn  Microsoft Windows netbios-ssn
445/tcp    open  microsoft-ds?
3306/tcp   open  mysql        MySQL (unauthorized)
Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 7.76 seconds

C:\Windows\System32>

```

Aggressive Scan:

 **Command used:** `nmap -A <IP>`

Findings:

- Collected detailed information on services.
- Identified possible security risks.
- OS detection results included.

```
C:\Windows\System32>nmap -A 19.
Starting Nmap 7.95 ( https://nmap.org ) at 2025-02-13 23:52 India Standard Time
Nmap scan report for 1.
Host is up (0.00076s latency).
Not shown: 996 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
135/tcp    open  msrpc        Microsoft Windows RPC
139/tcp    open  netbios-ssn  Microsoft Windows netbios-ssn
445/tcp    open  microsoft-ds?
3306/tcp   open  mysql        MySQL (unauthorized)
No exact OS matches for host (If you know what OS is running on it, see https://nmap.org/submit/ ).
TCP/IP fingerprint:
OS:SCAN(V=7.95%E=4%D=2/13%OT=135%CT=1%CU=40506%PV=Y%D=0%DC=L%G=Y%TM=67AE38
OS:A3%P=1686-pc-windows-windows)SEQ(SP=103%GCD=2%ISR=10B%TI=I%CI=I%II=I%SS=
OS:S%TS=A)SEQ(SP=104%GCD=1%ISR=10A%TI=I%CI=I%II=I%SS=S%TS=A)SEQ(SP=105%GCD=
OS:1%ISR=107%TI=I%CI=I%II=I%SS=S%TS=A)SEQ(SP=105%GCD=1%ISR=10B%TI=I%CI=I%II
OS:=I%SS=S%TS=A)SEQ(SP=107%GCD=1%ISR=109%TI=I%CI=I%II=I%SS=S%TS=A)OPS(O1=MF
OS:FD7NW8ST11%O2=MFFD7NW8ST11%O3=MFFD7NW8NNT11%O4=MFFD7NW8ST11%O5=MFFD7NW8S
OS:T11%O6=MFFD7ST11)WIN(W1=FFFF%W2=FFFF%W3=FFFF%W4=FFFF%W5=FFFF%W6=FFFF)ECN
OS:(R=Y%DF=Y%T=80%W=FFFF%O=MFFD7NW8NNS%CC=N%Q=)T1(R=Y%DF=Y%T=80%S=0%A=S+F=
OS:AS%RD=0%Q=)T2(R=Y%DF=Y%T=80%W=0%S=Z%A=S+F=AR%O=%RD=0%Q=)T3(R=Y%DF=Y%T=80
OS:%W=0%S=Z%A=0%F=AR%O=%RD=0%Q=)T4(R=Y%DF=Y%T=80%W=0%S=AA%O=F=R%O=%RD=0%Q=
OS:)T5(R=Y%DF=Y%T=80%W=0%S=Z%A=S+F=AR%O=%RD=0%Q=)T6(R=Y%DF=Y%T=80%W=0%S=AA
OS:A=0%F=R%O=%RD=0%Q=)T7(R=Y%DF=Y%T=80%W=0%S=Z%A=S+F=AR%O=%RD=0%Q=)U1(R=Y%
OS:DF=N%T=80%IPL=164%UN=0%RIPL=G%RID=G%RIPCK=Z%RUCK=G%RUD=G)IE(R=Y%DFI=N%T=
OS:80%CD=Z)

Network Distance: 0 hops
Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows
```

```
OS:80%CD=Z)

Network Distance: 0 hops
Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows

Host script results:
  smb2-time:
    date: 2025-02-13T18:23:18
  _ start_date: N/A
  smb2-security-mode:
    3:1:1:
      Message signing enabled but not required

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 32.87 seconds

C:\Windows\System32>
```

OS Detection Scan:

 **Command used:** `nmap -O <IP>`

Findings:

- Detected operating system details of the target.

```
nmap done: 1 IP address (1 host up) scanned in 32.87 seconds

C:\Windows\System32>nmap -O 192.168.1.100
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OS:SCAN(V=7.95%E=4%D=2/13%OT=135%CT=1%CU=30350%PV=Y%DS=0%DC=L%G=Y%TM=67AE39
OS:E1%P=1686-pc-windows-windows)SEQ(SP=102%GCD=1%ISR=109%TI=I%CI=I%II=I%SS=
OS:S%TS=A)SEQ(SP=103%GCD=1%ISR=107%TI=I%CI=I%II=I%SS=S%TS=A)SEQ(SP=105%GCD=
OS:1%ISR=108%TI=I%CI=I%II=I%SS=S%TS=A)SEQ(SP=105%GCD=3%ISR=100%TI=I%CI=I%II
OS:=I%SS=S%TS=A)SEQ(SP=107%GCD=1%ISR=108%TI=I%CI=I%II=I%SS=S%TS=A)OPS(O1=MF
OS:FD7NW8ST11%O2=MF7D7NW8ST11%O3=MF7D7NW8NNT11%O4=MF7D7NW8ST11%O5=MF7D7NW8S
OS:T11%O6=MF7D7ST11)WIN(W1=FFFF%W2=FFFF%W3=FFFF%W4=FFFF%W5=FFFF%W6=FFFF)ECN
OS:(R=Y%DF=Y%T=80%W=FFFF%O=MF7D7NW8NNS%CC=N%Q=)T1(R=Y%DF=Y%T=80%W=0%A=S+F=
OS:AS%RD=0%Q=)T2(R=Y%DF=Y%T=80%W=0%S=Z%A=S+F=AR%O=RD=0%Q=)T3(R=Y%DF=Y%T=80
OS:%W=0%S=Z%A=0F=AR%O=RD=0%Q=)T4(R=Y%DF=Y%T=80%W=0%S=A%O=RD=0%Q=)T5(R=Y%DF=Y%T=80%W=0%S=Z%A=S+F=AR%O=RD=0%Q=)T6(R=Y%DF=Y%T=80%W=0%S=A%
OS:A=0F=AR%O=RD=0%Q=)T7(R=Y%DF=Y%T=80%W=0%S=Z%A=S+F=AR%O=RD=0%Q=)U1(R=Y%
OS:DF=N%T=80%IPL=164%UN=0%RIPL=G%RID=G%RIPCK=Z%RUCK=G%RUD=G)IE(R=Y%DFI=N%T=
OS:80%CD=Z)
```

Observations and Recommendations

- The scanned system has common web services running (HTTP, HTTPS).
- No immediate vulnerabilities detected, but regular security patches are recommended.
- Secure SSH by restricting access and using key-based authentication.
- Firewall rules should be reviewed to ensure no unnecessary open ports.

⚠️ Security Issues Detected

1 🚫 HTTP Server Misconfiguration

- `http-methods:` `TRACE` method is enabled, which can be used for **Cross-Site Tracing (XST)** attacks.
- **Risk:** Attackers can steal cookies or session information.
- **Fix:** Disable `TRACE` in Apache configuration.

2 🚫 Expired SSL Certificate on HTTPS (Port 443)

- **Risk:** Users visiting the website may get security warnings. Attackers can intercept encrypted communication.
- **Fix:** Renew the SSL certificate.

3 🚫 SMTP Service Running on Port 25

- **Risk:** If misconfigured, it may allow **spam emails or email spoofing**.
- **Fix:** Use proper authentication and encryption (TLS/SSL).

4 🚫 TLS Randomness Warning (`_ssl-date: TLS randomness does not represent time`)

- **Risk:** Indicates an issue with how SSL/TLS randomness is generated, which could weaken encryption.
- **Fix:** Update OpenSSL and verify the system clock.

🏁 Conclusion

Network scanning with Nmap provided insights into the active hosts, open ports, and running services. The scan results highlight potential security measures that can be implemented to secure the network effectively.

📄 **Submitted By:** [Deepika Deshmukh]

📅 **Date:** 12th Feb

📁 **Submission:** WEE4 Day 1